

SolvencyInsight

AI-Powered Financial Distress Prediction System

Complete Technical Documentation

Team Member	Primary Responsibility
Rohan	Machine Learning Module & Backend Architecture
Jonathan	Machine Learning Module & Backend Development
Neha	Data Preparation & Frontend Development
Mariya	Frontend Development & UI/UX

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1. Executive Summary

1.1 What is SolvencyInsight?

SolvencyInsight is an enterprise-grade financial distress prediction platform that combines **traditional financial analysis** (Altman Z-Score) with **modern machine learning** (XGBoost with SHAP explainability) to predict corporate bankruptcy risk with **97.97% ROC-AUC accuracy**.

1.2 Problem Statement

Corporate bankruptcy can devastate stakeholders - employees lose jobs, investors lose capital, and suppliers lose revenue. Early detection of financial distress allows investors to divest, creditors to adjust terms, management to take corrective action, and regulators to intervene proactively.

1.3 Key Capabilities

Feature	Description
Single Company Analysis	Upload CSV or enter data manually for instant risk assessment
Bulk Analysis	Process hundreds of companies simultaneously
SHAP Explanations	Understand WHY a company is flagged as high-risk
Altman Z-Score	Classic bankruptcy predictor with zone classification
PDF Reports	Generate professional reports for stakeholders

1.4 Technology Stack

Layer	Technologies
Frontend	React 18, TypeScript, TailwindCSS, Recharts, Vite
Backend	FastAPI, Pydantic, Uvicorn, Python 3.11+
Machine Learning	XGBoost, SHAP, Scikit-learn, Pandas, NumPy
Data	10,000 Company Dataset, 30 Industries, 5 Company Sizes

2. Project Architecture

2.1 Directory Structure

```
insolvency-prevention-system/
|-- backend/ # FastAPI Backend
|   |-- app/
|       |-- main.py # API endpoints
|       |-- models/schemas.py # Pydantic models
|       |-- services/ # Business logic
|-- frontend/ # React Frontend
|   |-- src/
|       |-- components/ # UI components
|       |-- pages/ # Page components
|       |-- services/api.ts # API client
|-- ml_models/ # ML Module
|   |-- insolvency_predictor_v2.py # Enhanced model
|   |-- saved_models/ # Trained models
|-- data/training_companies_10k/ # Dataset
```

2.2 Data Flow

User uploads CSV --> FileUpload component --> api.ts (Axios) --> FastAPI endpoint --> Pandas reads CSV --> Feature Engineering (12 to 41 features) --> XGBoost prediction --> SHAP explanation --> JSON response --> React renders results

2.3 Key Integration Points

Vite Proxy: Frontend dev server (port 5173) proxies /api requests to backend (port 8001), enabling seamless development without CORS issues.

FormData Upload: CSV files sent as multipart/form-data, parsed by FastAPI using python-multipart.

Model Loading: On backend startup, the trained XGBoost model is loaded from pickle file into memory for fast inference.

3. Data Layer

3.1 Dataset Overview

Metric	Value
Total Companies	10,000
Healthy Companies	5,302 (53.0%)
Distressed Companies	4,698 (47.0%)
Industries	30
Company Sizes	5 categories (micro, small, medium, large, enterprise)
Economic Cycles	4 phases (expansion, peak, contraction, trough)

3.2 Input Features (12 Base Ratios)

Altman Z-Score Components (5 ratios):

Feature	Formula	Weight
working_capital_to_total_assets	$(CA - CL) / TA$	1.2
retained_earnings_to_total_assets	RE / TA	1.4
ebit_to_total_assets	$EBIT / TA$	3.3
market_value_equity_to_total_liabilities	MVE / TL	0.6
sales_to_total_assets	$Sales / TA$	1.0

Additional Financial Ratios (7 ratios):

Feature	Healthy Range	Distressed Range
current_ratio	> 1.5	< 1.0
quick_ratio	> 1.0	< 0.5
debt_to_equity	< 1.0	> 2.5
interest_coverage	> 3.0	< 1.5
net_profit_margin	> 5%	< 0%
return_on_assets	> 5%	< 0%
return_on_equity	> 10%	< 0%

4. Machine Learning Module

4.1 Model Selection: Why XGBoost?

Criterion	XGBoost V2	Random Forest	Logistic Regression
ROC-AUC	97.97%	97.60%	97.30%
Non-linearity	Excellent	Good	Limited
Feature Interactions	Automatic	Automatic	Manual only
Regularization	L1 + L2 + gamma	Tree constraints	L2 only
SHAP Support	Native	Supported	Coefficients

4.2 XGBoost Hyperparameters

Parameter	Value	Purpose
n_estimators	200	Number of boosting rounds
max_depth	6	Tree depth (captures interactions)
learning_rate	0.05	Step size shrinkage
reg_alpha	0.5	L1 regularization (Lasso)
reg_lambda	2.0	L2 regularization (Ridge)
subsample	0.8	Row sampling per tree
colsample_bytree	0.8	Feature sampling per tree
min_child_weight	5	Min samples per leaf
gamma	0.1	Min loss reduction for split
early_stopping_rounds	20	Stop if no improvement

4.3 Training Pipeline

1. **Extract 12 base features** from input DataFrame
2. **Engineer 41 total features** (see Section 8)
3. **Train-test split** (80/20, stratified by class)
4. **Train XGBoost** with early stopping on validation set
5. **Create SHAP TreeExplainer** for interpretability
6. **Calculate metrics** (accuracy, precision, recall, F1, ROC-AUC)
7. **5-fold cross-validation** for robust performance estimate
8. **Save model** to pickle file

4.4 SHAP Explainability

SHAP (SHapley Additive exPlanations) values measure each features contribution to the prediction:

Prediction = Base Value + SHAP(feature_1) + SHAP(feature_2) + ... + SHAP(feature_n)

Positive SHAP: Feature INCREASES bankruptcy risk (e.g., high debt)

Negative SHAP: Feature DECREASES bankruptcy risk (e.g., high current ratio)

Near Zero: Feature has minimal impact on this prediction

4.5 Altman Z-Score

Formula: $Z = 1.2 \cdot X_1 + 1.4 \cdot X_2 + 3.3 \cdot X_3 + 0.6 \cdot X_4 + 1.0 \cdot X_5$

Zone	Z-Score Range	Interpretation
Safe	$Z > 2.99$	Low bankruptcy probability
Grey	$1.81 \leq Z \leq 2.99$	Uncertain, monitoring needed
Distress	$Z < 1.81$	High bankruptcy probability

5. Model Comparison Analysis

5.1 Performance Results

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
XGBoost V2	91.15%	93.01%	87.77%	90.31%	97.97%
Random Forest	90.55%	93.51%	85.85%	89.52%	97.60%
Logistic Regression	90.15%	91.42%	87.23%	89.28%	97.30%

5.2 Cross-Validation Results (5-Fold)

Model	CV Mean	CV Std	95% Confidence Interval
XGBoost V2	98.14%	0.17%	[97.80%, 98.48%]
Random Forest	97.61%	0.28%	[97.06%, 98.17%]
Logistic Regression	97.32%	0.27%	[96.79%, 97.85%]

5.3 Why XGBoost V2 is Best

- Highest ROC-AUC (97.97%):** Best ability to distinguish healthy vs distressed companies across all thresholds.
- Feature Engineering:** 41 engineered features capture complex relationships (composite scores, interactions, zone indicators).
- Gradient Boosting:** Sequential error correction outperforms Random Forests bagging approach.
- Built-in Regularization:** L1 + L2 + gamma prevents overfitting on financial data with spurious correlations.
- Non-linear Patterns:** Captures thresholds (e.g., debt_to_equity > 3.0 = danger zone) that Logistic Regression misses.
- SHAP Integration:** TreeExplainer provides fast, accurate, auditable explanations for each prediction.

5.4 Confusion Matrix Analysis (XGBoost V2)

Metric	Value	Interpretation
True Negatives	998	Healthy companies correctly identified
True Positives	825	Distressed companies correctly identified
False Positives	62	Healthy companies incorrectly flagged (Type I)
False Negatives	115	Distressed companies missed (Type II)

Specificity	94.15%	True Negative Rate
Sensitivity	87.77%	True Positive Rate (Recall)

6. Feature Engineering Deep Dive

Feature engineering transforms **12 base features** into **41 total features** by creating derived metrics, indicators, and interaction terms.

6.1 Feature Categories

Category	Features	Purpose
Z-Score	altman_z_score, z_score_safe, z_score_grey, z_score_distress	Classic bankruptcy indicator
Liquidity	liquidity_gap, severe_liquidity_stress, moderate_liquidity_stress	Short-term solvency
Leverage	high_leverage, extreme_leverage, leverage_coverage_ratio	Debt risk indicators
Profitability	negative_profit, negative_roa, negative_roe, profitability_score	Profit health
Working Capital	negative_working_capital, working_capital_trend	WC health signals
Efficiency	asset_utilization, capital_efficiency	Asset productivity
Coverage	interest_coverage_safe, interest_coverage_danger	Debt service ability
Interactions	leverage_liquidity_interaction, profit_leverage_interaction, z_score_coverage_interaction	Coverage interaction
Composite	financial_distress_score, financial_health_score	Aggregated signals

6.2 Top 10 Most Important Features

Rank	Feature	Importance	Description
1	financial_distress_score	40.29%	Sum of 5 warning flags
2	profitability_score	13.79%	Average of profit margins
3	z_score_leverage_interaction	5.89%	Z-Score adjusted for debt
4	z_score_grey	4.55%	In grey zone indicator
5	negative_retained_earnings	2.65%	Accumulated losses flag
6	financial_health_score	2.55%	Sum of 5 health indicators
7	retained_earnings_to_total_assets	2.40%	Original Altman component
8	interest_coverage_safe	2.12%	Coverage > 3x indicator
9	altman_z_score	1.81%	Continuous Z-Score
10	interest_coverage_danger	1.54%	Coverage < 1.5x indicator

6.3 Key Engineered Features Explained

financial_distress_score (Most Important - 40.29%)

Combines 5 binary warning flags into a single 0-5 score:

```
financial_distress_score = (  
negative_working_capital + # WC/TA < 0  
severe_liquidity_stress + # Current Ratio < 1.0  
high_leverage + # D/E > 2.0  
negative_profit + # Net Margin < 0  
interest_coverage_danger # Coverage < 1.5x  
)
```

Score 0 = No red flags (healthy), Score 5 = Severe distress

profitability_score (Second Most Important - 13.79%)

Average of three profitability metrics:

```
profitability_score = (net_profit_margin + return_on_assets + return_on_equity) / 3
```

z_score_leverage_interaction (Third Most Important - 5.89%)

Adjusts Z-Score for leverage level:

```
z_score_leverage_interaction = altman_z_score / (debt_to_equity + 0.5)
```

A company with high Z-Score but also high debt is riskier than Z-Score alone suggests.

7. Team Contributions

7.1 Rohan & Jonathan: ML Module & Backend

Responsibilities:

XGBoost model architecture and hyperparameter tuning

Feature engineering (12 to 41 features)

SHAP explainability integration

Altman Z-Score calculation and zone classification

FastAPI backend development

Pydantic schema validation

Model training, evaluation, and comparison

Key Files: `insolvency_predictor_v2.py`, `main.py`, `schemas.py`, `train_model_v2.py`, `model_comparison_v2.py`

7.2 Neha: Data Preparation & Frontend

Responsibilities:

Dataset generation (10,000 companies across 30 industries)

Industry-specific financial characteristics

Economic cycle simulation

CSV parsing and validation

API service layer (`api.ts` with Axios)

FileUpload component (drag-and-drop)

DataTable component (sortable, searchable)

Key Files: `generate_large_dataset.py`, `api.ts`, `FileUpload.tsx`, `DataTable.tsx`

7.3 Mariya: Frontend Development & UI/UX

Responsibilities:

React page components

InsolvencyAnalysis page (main analysis interface)

SHAP visualization using Recharts

RiskGauge component (circular probability display)

TailwindCSS styling and dark theme

Responsive design for all screen sizes

Toast notification system

Key Files: `InsolvencyAnalysis.tsx`, `RiskGauge.tsx`, `ToastContext.tsx`, `tailwind.config.js`

7.4 All Team Members: Integration

Frontend-backend integration via Vite proxy

End-to-end testing

Error handling and user feedback

Performance optimization

Documentation

8. Conclusion

SolvencyInsight demonstrates the practical application of machine learning to financial risk assessment. The system combines:

Traditional Finance: Altman Z-Score provides an interpretable, academically-validated baseline that financial professionals trust.

Modern ML: XGBoost captures non-linear patterns in financial data that traditional methods miss, achieving 97.97% ROC-AUC.

Explainable AI: SHAP values make predictions transparent and auditable, critical for regulatory compliance.

User-Friendly Interface: React frontend makes complex analysis accessible to non-technical users.

Key Achievements:

Metric	Value
Model Accuracy	91.15%
ROC-AUC Score	97.97%
Cross-Validation Mean	98.14%
Dataset Size	10,000 companies
Engineered Features	41
Industries Covered	30

RECOMMENDATION: Deploy XGBoost V2 with feature engineering for production use. The models combination of high accuracy, robust cross-validation performance, built-in regularization, and SHAP explainability makes it ideal for financial distress prediction applications.